

That way, no one will confuse the binary number 10000 (the number 16 in decimal) with the decimal number “ten thousand.” The numeral systems program removes these prefixes before displaying the number.

```
1. """Numeral System Counters, by Al Sweigart al@inventwithpython.com
2. Shows equivalent numbers in decimal, hexadecimal, and binary.
3. View this code at https://nostarch.com/big-book-small-python-projects
4. Tags: tiny, math"""
5.
6.
7. print('Numeral System Counters, by Al Sweigart al@inventwithpython.com
8.
9. This program shows you equivalent numbers in decimal (base 10),
10. hexadecimal (base 16), and binary (base 2) numeral systems.
11.
12. (Ctrl-C to quit.)
13. '''
14.
15. while True:
16.     response = input('Enter the starting number (e.g. 0) > ')
17.     if response == '':
18.         response = '0' # Start at 0 by default.
19.         break
20.     if response.isdecimal():
21.         break
22.     print('Please enter a number greater than or equal to 0.')
23. start = int(response)
24.
25. while True:
26.     response = input('Enter how many numbers to display (e.g. 1000) > ')
27.     if response == '':
28.         response = '1000' # Display 1000 numbers by default.
29.         break
30.     if response.isdecimal():
31.         break
32.     print('Please enter a number.')
33. amount = int(response)
34.
35. for number in range(start, start + amount): # Main program loop.
36.     # Convert to hexadecimal/binary and remove the prefix:
37.     hexNumber = hex(number)[2:].upper()
38.     binNumber = bin(number)[2:]
39.
40.     print('DEC:', number, '    HEX:', hexNumber, '    BIN:', binNumber)
```

After entering the source code and running it a few times, try making experimental changes to it. On your own, you can also try to figure out how to do the following:

- Enter a new row for *octal*, the base-8 number system, using Python’s `oct()` function.
- Search the internet for “number system conversion” to find out how to implement your own `bin()`, `oct()`, and `hex()` functions.